Sibley School of Mechanical and Aerospace Engineering

MAE 3240 Heat Transfer

Spring 2026

Problem Set #5

Posted: Monday, Mar 2 **Due: Tuesday, Mar 10, 11:59 pm**

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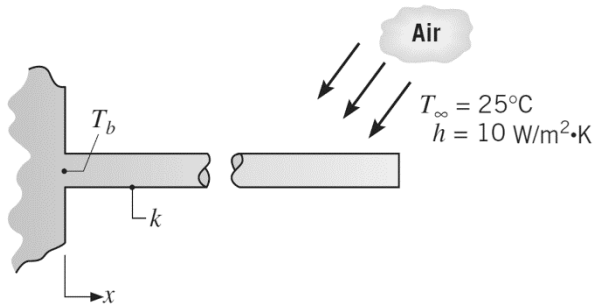
1. (60 pt) A cylindrical pin fin with a radius of r = 1 mm and length of L = 5 cm is exposed to airflow with a temperature of T_∞ = 25 °C and a heat transfer coefficient of h = 10 W/m^2·K. Thermal conductivity of the pin fin is k = 200 W/m·K. The tip of the pin fin is thermally insulated. Use the fin problem solving workflow discussed in the lecture to solve the following questions.



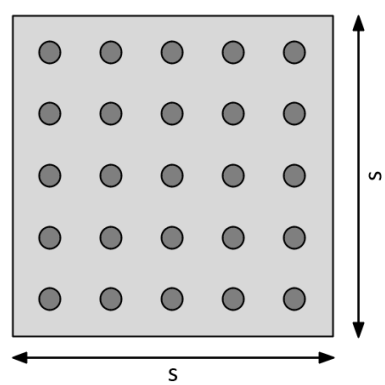
   (a) If the fin base temperature is T_b = 125 °C, please calculate the fin temperature at $x = L/2$ and $x = L$.

   (b) Calculate the fin effectiveness ε_f and the fin efficiency η_f based on their definition and results provided by Table 3.4 of the textbook.

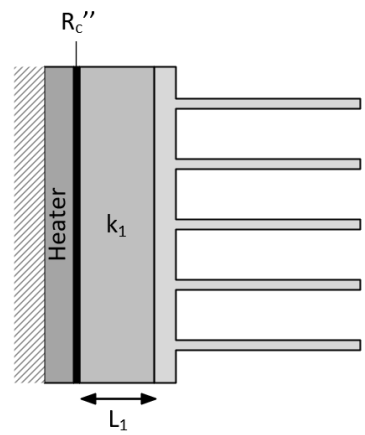
   (c) Calculate the fin efficiency η_f based on results provided by Table 3.5. Compare the results of (b) and (c) and discuss if Table 3.5 provided a good approximation to the fin efficiency.

   (d) Calculate the thermal resistance of this pin fin R_f.

   (e) The pin fins then form a 5×5 array mounted on a base plate with a side length of s = 3 cm, as shown in the figure on the right-hand side. Calculate the overall efficiency of the fin array η_o and the thermal resistance of the fin array R_o. Calculate how much heat is dissipated through the unfined surface and how much is through the fins.



(f) The heat sink consisting of this fin array is used to improve the heat dissipation from a thin film heater, as shown in the figure on the right-hand side. The heat sink is mounted on a slab with thickness L_1 = 2 cm and thermal conductivity k_1 = 10 W/m·K. There is a thermal adhesive between the thin film heater and the slab, resulting in a contact thermal resistance R_c = 10^{-4} m^2·K/W. Thermal insulation was applied to the backside of the heater. The heater generates a uniform heat flux q'' = 1000 W/m^2, which was dissipated through the heat sink. The temperature across the thin film heater is uniform. Thermal resistance for the base of the heat sink can be neglected. There is no contact thermal resistance between the base of the heat sink and the slab. Please draw the thermal resistance network of this system and calculate the temperature of the thin film heater T_h. Compared with the condition without the heat sink, *i.e.*, directly exposing the heater and the slab to ambient air, how much temperature reduction of the heater is achieved?



2. (30 pt) A long wire of diameter D = 1 mm is submerged into an oil bath of temperature T_∞ = 20 °C. Density ρ, thermal conductivity k, and heat capacity c of the wire are 4000 kg/m^3, 400 W/m·K, and 500 J/kg·K, respectively. The wire has an electrical resistance per unit length of R_e' = 0.01 Ω/m. The oil bath can dissipate heat with a heat transfer coefficient of h = 500 W/m^2·K. At the initial moment t = 0 s, a current of I = 100 A flows through the wire and produces Joule heating.
   (a) Calculate the steady-state temperature of the wire.
   (b) Apply the lumped capacitance method to derive an expression for the wire temperature as a function of time. How long does it take for the wire to reach a temperature that is within 5 °C of the steady-state temperature?

3. (10 pt) Complete the portfolio assignment.
   Please select one homework assignment from this course that you found most valuable or educational. Write a 1-2 paragraph reflection discussing what you learned from that assignment

Heat Transfer Problem Set 5

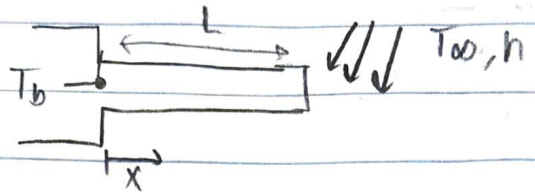
Problem 1

given: $r = 1\,mm$ $L = 5\,cm$ $T_\infty = 25°C$ $h = 10\,W/m^2 \cdot K$
$k = 200\,W/m \cdot K$, thermally insulated tip

find: $T(L/2)$, $T(L)$, ϵ_f, η_f, R_f, η_o, R_o, q_{uf}, q_f

assume: 1D heat conduction

schematic:



analysis

a) $A_c = \pi r^2 = \pi(1\times10^{-3}m)^2 = \pi\times10^{-6}\,m^2$

$P = 2\pi r = 2\pi(1\times10^{-3})m = 2\pi\times10^{-3}\,m$

$$m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{(10)(2\pi\times10^{-3})}{(200)(\pi\times10^{-6})}} = 10$$

for thermally insulated tip:

$$\frac{\theta(x)}{\theta_b} = \frac{\cosh(m(L-x))}{\cosh(mL)}$$

so $\theta(L) = \theta_b \dfrac{\cosh(0)}{\cosh(mL)} = \dfrac{\theta_b}{\cosh(mL)} = \dfrac{T_b - T_\infty}{\cosh(mL)}$

$$\theta(L) = \frac{125-25}{\cosh(10\times5\times10^{-2})} = 88.68\,°C$$

$$T(L) - T_\infty = \theta(L) \Rightarrow \boxed{T(L) = 113.68°C}$$

similarly $\theta(L/2) = \dfrac{\cosh(mL/2)}{\cosh(mL)}\theta_b = (125-25)\dfrac{\cosh(10\times5\times10^{-2}/2)}{\cosh(10\times5\times10^{-2})}$

$$\theta(L/2) = 91.46°C \Rightarrow \boxed{T(L/2) = \theta(L/2) + T_\infty = 116.46°C}$$

b) $\varepsilon_f = q_f / q_{nf}$

From table 3.4 : $q_f = M \tanh(mL)$

where $M = \sqrt{hPkA_c}\,\theta_b = \pi/5 = 0.628$

so $q_f = \frac{\pi}{5} \tanh(10 \times 5 \times 10^{-2}) = 0.290\ W$

and $q_{nf} = hA_c(T_b - T_\infty) = (10)(\pi \times 10^{-6})(100) = 3.14 \times 10^{-3}\ W$

so $\boxed{\varepsilon_f = q_f / q_{nf} = 92.42}$

$$\eta_f = \frac{q_f}{q_{max}} = \frac{M \tanh(mL)}{hA_f(T_b - T_\infty)} = \frac{\pi/5 \tanh(10 \times 5 \times 10^{-2})}{10(2\pi \times 10^{-3})(5 \times 10^{-2})(100)}$$

$$\boxed{\eta_f = 0.924}$$

c) From table 3-5 :

For pin fins $\eta_f = \frac{\tanh mL_c}{mL_c}$ $\quad L_c = L + D/4$

$= 5 \times 10^{-2} m + \frac{2 \times 10^{-3}}{4}$

$= \frac{\tanh(10 \times 0.0505)}{10 \times 0.0505}$ $\quad = 0.0505 m$

$\boxed{\eta_f = 0.923}$ This is very close to our previously calculated fin efficiency. (0.1% diff)

d) $$R_f = \frac{1}{\eta_f h A_f} = \frac{1}{(0.924)(10)(2\pi \times 10^{-3})(5 \times 10^{-2})}$$

$$\boxed{R_f = 344.5\ K/W}$$

e) $A_f = PL = (2\pi \times 10^{-3})(5 \times 10^{-2}) = 10\pi \times 10^{-5}\ m^2$

$A_{uf} = A - NA_s = (3 \times 10^{-2})^2 - 25(\pi \times 1 \times 10^{-3})^2 = 6.53 \times 10^{-4} m^2$

$A_{tot} = A_{uf} + NA_f = 8.50 \times 10^{-3} m^2$

so $\eta_o = 1 - \frac{N A_f}{A_{tot}}(1-\eta_f)$

$$= 1 - \frac{25 \times 10\pi \times 10^{-5}}{8.5 \times 10^{-3}}(1-0.924)$$

$$\boxed{\eta_o = 0.930}$$

$$R_o = \frac{1}{\eta_o h A_{tot}} = \frac{1}{10 \times 0.930 \times 8.5 \times 10^{-3}} = \boxed{12.6\ K/W}$$

f)

$R_c = 10^{-4}/(3\times10^{-2})^2 = 1/9\ K/W$ $\quad R_{cond} = \frac{L}{k_1 A} = 20/9\ K/W$

$R_o = 12.6\ K/W$

$R_{tot} = \Sigma R = 14.93\ K/W$

$\Delta T = q R_{tot} \Rightarrow T_h = T_\infty + (3\times10^{-2})^2(1000)(14.93)$

$\Rightarrow T_h = 311.58\ K = \boxed{38.4°C}$

<u>without the heat sink</u>:

$R''_{conv} = 1/h = 0.1\ m^2 \cdot K/W$ $\quad R''_{cond} = L/k = 2\times10^{-3}\ m^2 K/W$

$R''_{tot} = \Sigma R = 0.1021\ m^2 K/W$

$T_h = T_\infty + q'' R''_{tot} = 298.15 + (1000)(0.1021)$

$T_h = 400.25\ K = \boxed{127.1°C}$

The heat sink reduces the temperature by

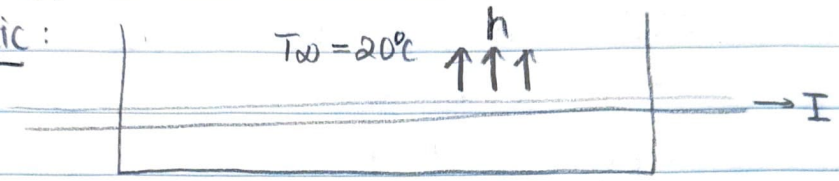
$\boxed{\Delta T = 88.7°C}$ → significant change

Problem 2

given: $D = 1mm$, $T_\infty = 20°C$, $\rho = 4000\ Kg/m^3$, $K = 400\ W/m \cdot K$
$C = 500\ J/Kg \cdot K$, $R_e' = 0.01\ \Omega/m$, $h = 500\ W/m^2 \cdot K$
at $t = 0$, $I = 100\ A$

find: T_{ss}, $T(t)$

schematic:



analysis

a) energy balance: $\dot{E}_{in} - \dot{E}_{out} + \dot{E}_g = \dot{E}_{st}$

$$\Rightarrow \dot{E}_g = \dot{E}_{out} \Rightarrow I^2 R_e' L = hA(T_{ss} - T_\infty)$$

$$\Rightarrow I^2 R_e' = hP(T_{ss} - T_\infty)$$

$$\Rightarrow T_{ss} = T_\infty + \frac{I^2 R_e'}{hP} = 20°C + \frac{100^2 (0.01)}{500\pi(1\times10^{-3})} = \boxed{83.66°C}$$

b) Transient energy balance:

$$\dot{E}_{in} - \dot{E}_{out} + \dot{E}_g = \dot{E}_{st}$$

$$-hA(T - T_\infty) + I^2 R_e' L = \rho V C\, dT/dt$$

$$-hP\theta + I^2 R_e' = \rho A C\, d\theta/dt$$

at steady state $I^2 R_e' = hP(T_{ss} - T_\infty)$

$$-hP\theta + hP(T_{ss} - T_\infty) = \rho A C\, d\theta/dt$$

$$hP(T_{ss} - T_\infty - T + T_\infty) = \rho A C\, d\theta/dt$$

$$hP(T_{ss} - T) = \rho A C\, d\theta/dt \qquad \theta = T - T_{ss}$$

$$-hP\theta = \rho A C\, d\theta/dt \Rightarrow -\frac{hP}{\rho A C} dt = \frac{1}{\theta} d\theta$$

$$\Rightarrow \ln(\theta) = -\frac{h\pi D}{\rho \frac{\pi D^2}{4} C} = -\frac{4h}{\rho D C} \Rightarrow \theta(t) = \theta_i \exp\left(-\frac{4h}{\rho D C}\right)$$

$$\Rightarrow T(t) - T_{ss} = (T_i - T_{ss}) \exp\left(-\frac{4h}{\rho D C}\right)$$

$$T(t) = T_{ss} + (T_i - T_{ss}) \exp\left(\frac{-4h}{\rho D C}\right)$$

plugging in constants:

$$T(t) = 83.66°C + (20 - 83.66°C) \exp(-t)$$

$$\boxed{T(t) = 83.66 - 63.66 \exp(-t)} \quad (°C)$$

$$T(t^*) = 78.66 = 83.66 - 63.66 \exp(-t^*)$$

$$-5 = -63.66 \exp(-t)$$

$$\exp(-t) = 0.0785$$

$$\boxed{t = -\ln(0.0785) = 2.54 \text{ sec}}$$